

Linus Chirchir

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[ORCID](#) | [Google Scholar](#) | [Scopus](#) | [Web of Science](#) | [Professional Blog](#)

Profile

Population and Health Data Scientist and Doctoral Researcher specialising in the preparation, linkage and analysis of large-scale administrative datasets across health and social care research. Experienced in UK Trusted Research Environments, with research interests spanning population data science, health workforce research, data linkage and synthetic data.

Employment

Data Analyst, [Research Data Scotland](#), Edinburgh, United Kingdom **09/2024 - Present**

- Establishing standards for data cleaning, quality assurance, metadata, and documentation to enable secure and FAIR access for demographic and social science research.
- Process sensitive datasets for the Scottish National Safe Haven as part of UKRI Administrative Data Research programme
- Develop and evaluate synthetic datasets using statistical modelling approaches (e.g., regression-based and tree-based methods), assessing fidelity, utility, and disclosure risk

Health Data Analyst, [University of Edinburgh](#), Edinburgh, United Kingdom **09/2023 - 06/2024**

- Designed reproducible R and SQL data pipelines to integrate over 13,000 linked NHS orthopaedic records using regression modelling (linear and logistic) and record linkage methods to examine treatment pathways and patient outcomes
- Developed interactive Shiny dashboard for orthopaedic outcomes to support clinical reporting and evidence-based decision making.
- Coordinated data governance processes, metadata standardisation and data dictionaries with the Data Controller and Research Database Manager for audit purposes and reuse of data.

ICT Officer/Researcher, [Moi Teaching & Referral Hospital](#), Eldoret, Kenya **07/2008 - 12/2021**

- Led data quality improvement and analytics initiatives across 104 administrative health datasets
- Applied descriptive, correlation, regression, time-series analysis, and hypothesis testing to examine patterns in patient flow, service utilisation, workforce performance, trends in service activity, patient outcomes, and workforce metrics
- Developed key performance indicators (KPIs) with clinical and administrative workforce to support healthcare delivery, resource allocation, and performance monitoring.
- Contributed to empirical research resulting in nine peer-reviewed publications

Education

PhD in Health and Social Care, [Edinburgh Napier University](#), United Kingdom **10/2026 - Present**

Research Project: Using Nationally Representative Data to Understand the Nursing and Midwifery Professions in the United Kingdom

- Using nationally representative and linked data to investigate the nursing and midwifery workforce in the UK, with a focus on workforce characteristics, patterns and inequalities.

MSc in Health Data Science, [Swansea University](#), United Kingdom **01/2022 - 05/2023**

Dissertation: Evaluating Cognitive Assessments in the Progression to Early-Stage Alzheimer's Disease Using Machine Learning

- Developed and validated six machine learning models in R (Logistic Regression, KNN, SVM, Random Forest, GBM, AdaBoost) using the OASIS-3 dataset to predict early-stage Alzheimer's disease, achieving 91% accuracy and AUROC of 0.94.

Master of Business Administration (Health Leadership and Management), [United States International University-Africa](#), Kenya **09/2018 - 08/2019**

Project: Factors Influencing Electronic Health Record (EHR) Adoption by Nurses at Moi Teaching and Referral Hospital

- Conducted a quantitative cross-sectional study using survey data analysed in SPSS, applying correlation and linear regression techniques to assess determinants of EHR adoption to provide recommendations to support digital health implementation, including workforce training.

Master of Information Systems, [Kisii University](#), Kenya

01/2014 - 12/2015

Thesis: Effects of Health Information Systems on User Performance: A Survey of Health Organisations in Uasin Gishu County, Kenya

- Applied descriptive statistics, correlation analysis, multivariable regression modelling, and hypothesis testing to examine direct and mediated relationships between system characteristics and user performance.

Bachelor of Business Management, [Moi University](#), Kenya

09/2008 - 12/2012

- Developed foundational training in statistics, database systems, programming, and operations research, supporting later work in data management, statistical analysis, and health data science.

Grant Funding

Principal Investigator - £64,508.43

- Awarded funding through the Careers and Skills for Data-driven Research (CaSDaR) Network+, UKRI Research Technical Professional Network+ Call (Grant No. UKRI739), March 2026.

Honours and Awards

UKRI-Endorsed Global Talent, 2026

- Recognised through the UKRI Endorsed Funder route for professional contribution to secure administrative data research, research methodology and responsible data use.

Technical Skills

- **Programming and Statistical Analysis:** R, SQL, SPSS, Advanced Excel
- **Statistical Methods:** Regression modelling (linear, logistic, Poisson, negative binomial, Cox proportional hazards, and multilevel/mixed effects), hypothesis testing, correlation and chi-square analysis, longitudinal and cohort analysis, epidemiological analysis, causal inference (including propensity score methods), machine learning, and record linkage.
- **Data Visualisation and GIS:** R, Tableau, Power BI, QGIS, ArcGIS
- **Reproducible Research and Version Control:** Quarto, Git, GitHub

Peer Review and Research Engagement

- **ESRC Peer Review College**
Expert reviewer for population methods, data science and social statistics proposals
- **Peer Reviewer**
Reviewed 19 manuscripts across 7 journals, including the International Journal of Population Data Science, Journal of Medical Internet Research (JMIR), JMIR Medical Informatics, JMIR Bioinformatics and Biotechnology, JMIR Perioperative Medicine and PLOS ONE.
- **Mentor, The Mentoring Club**
Provide one-to-one mentoring to early-career professionals in data science and analytics, offering guidance on research methods, career development and professional practice.
- **Publons Academy Supervisor (Peer Review Mentor)**
Mentored 6 trainees on the Publons Practical Peer Review Training Course, providing structured feedback and final review assessments.

Certifications and Training

- EPSRC and ESRC Peer Review Training, November 2025
- Office for National Statistics Safe Researcher Training (ONS-SRT), June 2025
- Certified Publons Academy Peer Reviewer (September 2020) and Mentor (April 2021)

Memberships and Fellowships

- Research Data Alliance (RDA) (Member)
- The Federation for Informatics Professionals (Advanced Practitioner)
- Royal Statistical Society (Member and Graduate Statistician)
- Association of Professional Healthcare Analysts (Member)

Publications

Peer Reviewed Conferences

- **Chirchir, L.**, (2025). Automating Low-Fidelity Synthetic Data Generation from Metadata: A Reproducible Approach Using R. *International Journal of Population Data Science*, 10(4). <https://doi.org/10.23889/ijpds.v10i4.3134>
- Bishop, J., **Chirchir, L.**, Coey, R., Dunlop, A., Evans, H., Grzybowski, D., Harbinson, A., Magder, C., McCullough, N., McFerran, R., Muir, J., Ramage, A., Telford, S., Thomas, R. and Vilcan, T. (2025). ADRUK Ingest Improvement Task and Finish Group: Smoothing the ingest of data. *International Journal of Population Data Science*, 10(4). <https://doi.org/10.23889/ijpds.v10i4.3130>
- **Chirchir, L. K.**, Aruasa, W. K., Kugo, L. C., Rop, C. C., & Mwogi, T. S. Investigating the Use of Facebook by Medical Institutions in Kenya. *PACIS 2021 Proceedings*. 230. <https://aisel.aisnet.org/pacis2021/230/>
- **Chirchir, L. K.**, Aruasa, W. K., & Chebon, S. K. (2021). Change Process Factors Influencing Electronic Health Records Adoption by Nurses at Moi Teaching and Referral Hospital, Kenya. *Procedia Computer Science*, 181, 427–433. <https://doi.org/10.1016/j.procs.2021.01.187>
- **Chirchir, L. K.**, Aruasa, W. K., & Chebon, S. K. (2020). Determining Role of Social Factors in Adoption of Electronic Health Records by Nurses. *Studies in Health Technology and Informatics*, 272, 111–114. <https://doi.org/10.3233/SHTI200506>

Peer Reviewed Journal Articles

- **Chirchir, L. K.**, Aruasa, W. K., & Chebon, S. K. (2021). Technical Factors Influencing Electronic Health Records Adoption by Nurses at Moi Teaching and Referral Hospital, Kenya. *African Journal of Science, Technology, Innovation and Development*, 13(5), 551 – 558. <https://doi.org/10.1080/20421338.2020.1777676>
- Aruasa, W. K., **Chirchir, L. K.**, & Chebon, S. K. (2019). Implications of Physicians and Nurses' Professional Satisfaction on Patient Care. *International Journal of Public Health Science (IJPHS)*, 8 (3), 300 – 306. <http://doi.org/10.11591/ijphs.v8i3.20238>
- Chebon, S. K., Aruasa, W. K., & **Chirchir, L. K.** (2019). Effect of Inspirational Motivation and Idealized Influence on Employee Performance at Moi Teaching and Referral Hospital, Eldoret, Kenya. *International Journal of Business and Social Science*, 10(7), 131 – 140. <http://doi.org/10.30845/ijbss.v10n7p14>
- Aruasa, W. K., **Chirchir, L. K.**, & Chebon, S. K. (2019). Determinants of Physicians and Nurses' Professional Satisfaction at the Moi Teaching and Referral Hospital, Eldoret, Kenya. *Journal of Health, Medicine and Nursing*, 64, 6 – 17. <http://doi.org/10.7176/JHMN/64-02>
- Aruasa, W. K., **Chirchir, L. K.**, & Chebon, S. K. (2019). [Strategies for Improving Physicians and Nurses' Professional Satisfaction at the Moi Teaching and Referral Hospital, Eldoret, Kenya](#). *The Strategic Journal of Business & Change Management*, 6 (3), 216 – 226.
- Chebon, S. K., Aruasa, W. K., & **Chirchir, L. K.** (2019). [Influence of Individualized Consideration and Intellectual Stimulation on Employee Performance: Lessons from Moi Teaching and Referral Hospital, Eldoret, Kenya](#). *IOSR Journal of Humanities and Social Science*, 24 (7), 11 – 22.
- **Chirchir, L. K.**, Aruasa, W. K., & Chebon, S. K. (2019). [Perceived Usefulness and Ease of Use as Mediators of the Effect of Health Information Systems on User Performance](#). *European Journal of Computer Science and Information Technology*, 7 (1), 22 – 37.
- Ireri, E. N., Omondi, Y. A., **Chirchir, L. K.**, & Wafula, M. K. (2015). [Adoption of Enterprise Resource Planning System in Kenya: A Case Study of Kenya Ordnance Factories Corporation](#). *International Journal of Current Research*, 7 (4), 14702 – 14711.